

Martingales in Practice, Part I: When Is a Process Really a Martingale?

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Martingales sit at the center of derivative pricing, stochastic modeling, measure changes, and quantitative model validation, yet they are often recognized through incomplete shortcuts.

Abstract

This first article in a three-part series develops a practical three-route framework: direct conditional expectation, inspection of the finite-variation term in an Itô decomposition, and the square-integrable Brownian stochastic-integral theorem. It separates true martingales from local martingales and applies the framework to Brownian motion, squared Brownian motion, and compensated squared Brownian motion. The examples show why a flat simulated mean does not establish a martingale and why zero Itô drift initially yields only a local-martingale conclusion. The central message is practical: the probability measure, filtration and integrability argument must be made explicit before a martingale claim is used in pricing or model validation. This is the first part of a three part series.

Keywords

Martingale, local martingale, Brownian motion, Itô calculus, conditional expectation, compensation, quantitative model validation

1 Why martingales still cause errors in quantitative work

Suppose a simulated discounted asset price has an almost flat cross-sectional mean. Has its martingale property been validated?

Suppose an application of Itô's formula produces no visible dt -term. Is the process necessarily a true martingale?

Suppose that, for each fixed maturity, a random variable can be rewritten as a Brownian stochastic integral. Does the entire time-indexed process become a martingale?

Suppose two processes have the same mean and variance at every time. Must they have the same conditional dynamics?

The answer to all four questions is no. These distinctions are elementary in principle, but they remain important in practical quantitative work because the words *martingale*, *driftless process*, *stochastic integral* and *fair game* are often used as though they are interchangeable.

A martingale is not defined by a visually trendless sample path, a zero unconditional mean, or the presence of a dW_t -term. It is defined relative to a probability measure and an information filtration. For an integrable adapted process X , the essential requirement is

$$E[X_t | \mathcal{F}_u] = X_u, \quad 0 \leq u \leq t.$$

This equality is stronger than

$$E[X_t] = E[X_u].$$

The latter follows from the martingale property by the tower rule, but it cannot generally be reversed. A process may have zero mean at every horizon while possessing a nonzero, state-dependent conditional drift.

The distinction is not academic. In a Monte Carlo pricing engine, an unconditional-mean check may fail to detect an implementation whose conditional error changes sign across states. In an SDE derivation, a vanishing dt -term may establish only a local martingale, while valuation or measure-change arguments require a true martingale. In a density-process implementation, failure to verify this distinction can invalidate the proposed probability measure. In a model-validation exercise, confusing a time integral with a stochastic integral can turn an accumulated drift term into an incorrectly labeled source of innovation.

The modern role of martingales in finance developed through several related strands. Bachelier (1900) modeled speculative price movements using continuous random fluctuations. Samuelson (1965) related properly anticipated prices to fair-game reasoning. Itô (1944) introduced the stochastic integral that makes continuous-time martingale calculations possible. Black and Scholes (1973) and Merton (1973) used stochastic calculus and dynamic replication to derive option-pricing results. Harrison and Kreps (1979) and Harrison and Pliska (1981) formalized the connection between no-arbitrage, stochastic integrals and martingale measures, while Delbaen and Schachermayer (1994) established a general version of the fundamental theorem of asset pricing.

These developments created two complementary uses of martingales. The first is probabilistic: martingales describe processes whose conditional expectation does not change when information is updated. The second is financial: under a suitable pricing measure and numéraire, discounted tradable prices should possess a martingale or local-martingale structure.

The second use does not eliminate the first. It makes precise conditional reasoning more important. A pricing process is not “a martingale” without qualification. One should ask:

- under which probability measure;
- relative to which filtration;
- after which discounting or numéraire transformation;
- whether the result is local or true; and
- which integrability condition supports the conclusion.

This three-part series organizes the relevant reasoning into three routes.

The first route applies the conditional-expectation definition directly. It is the most general and logically decisive method.

The second route derives an Itô decomposition and examines the finite-variation drift. It is usually the fastest method in continuous-time models, but zero drift initially produces a local-martingale conclusion.

The third route applies a specialized theorem to processes already represented as Brownian stochastic integrals. If the integrand is predictable and square-integrable, the resulting process is a square-integrable martingale. This is not a third definition of martingale and does not apply directly to an ordinary ds -integral.

The series makes five practical contributions.

First, it separates a definition, a semimartingale diagnostic and a specialized stochastic-integral theorem that are often conflated.

Second, it preserves parallel proofs. Whenever feasible, the same process is examined through conditional expectations and its differential decomposition. This makes the relationship between predictable conditional growth and the dt -term explicit.

Third, it compares processes that are deceptively similar:

$$\int_0^t W_s dW_s \text{ and } \int_0^t W_s ds,$$

as well as

$$e^{\theta W_t} \text{ and } e^{\theta W_t - \theta^2 t/2}.$$

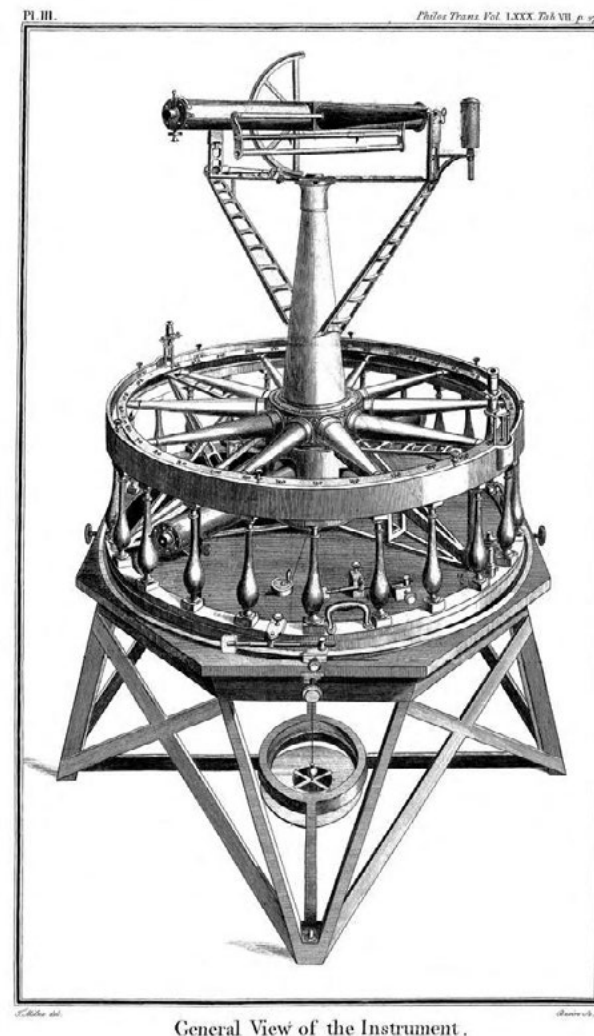
Fourth, it gives special attention to the difference among a local martingale, a true martingale, and a square-integrable martingale. This distinction matters for

stochastic exponentials, pricing-measure density processes, financial bubbles, and model validation.

Fifth, it supplies a reproducible Python laboratory¹. The simulations are not offered as proofs. They demonstrate what unconditional diagnostics can detect, what they can miss, and how a simple conditional-bin experiment can approximate the defining martingale restriction.

The mathematical treatment follows Doob (1953), Chung and Williams (1990), Williams (1991), Karatzas and Shreve (1991), Revuz and Yor (1999), Øksendal (2003), Protter (2005), and Klebaner (2012). Financial interpretations follow Shreve (2004) and Björk (2009), while the computational implementation is informed by Kloeden and Platen (1992), Higham (2001), and Glasserman (2004).

This martingale topic is presented as a three-part series.² Part I, presented here, establishes the three recognition routes, clarifies local versus true martingales, and applies the framework to Brownian motion, squared Brownian motion and its compensated form. Part II compares stochastic and ordinary time integrals, product processes, and variance identities. Part III develops exponential compensation, pricing-measure applications, the reproducible diagnostic laboratory, and the practitioner checklist.



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2 Three routes for recognizing a martingale

2.1 Information, integrability and the formal definition

Let

$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$

be a filtered probability space satisfying the usual conditions. The filtration

$\{\mathcal{F}_t\}_{t \geq 0}$

represents information accumulated through time. A process $X = \{X_t\}_{t \geq 0}$ is a martingale with respect to P and $\{\mathcal{F}_t\}$ when:

X_t is $\mathcal{F}_t$ -measurable,
 $|X_t| < \infty$,

and

$E[X_t | \mathcal{F}_u] = X_u, \quad 0 \leq u \leq t.$

The first condition is adaptedness. It prevents X_t from using information unavailable at time t .

The second condition is integrability. Without it, the ordinary conditional expectation in the definition need not be well defined.

The third condition is the substantive fair-game restriction. Once the information at time u is known, the expected value of the future process equals its current value.

By the tower property,

$E[X_t] = E[E[X_t | \mathcal{F}_u]] = E[X_u].$

Therefore, an integrable martingale has a constant unconditional mean. The converse is false.

For example,

$E[tW_t] = 0$

for every t , but tW_t is not a martingale. Similarly,

$E\left[\int_0^t W_s \, ds\right] = 0,$

yet the ordinary integrated Brownian-motion process is not a martingale.

This is the first practical warning:

$E[X_t] = E[X_u]$ is necessary but generally not sufficient.

Table 1 summarizes the three recognition routes without placing long formulas inside the table. The exact mathematical conditions are stated immediately afterward.

2.2 Route I: Direct conditional expectation

Let $W = \{W_t\}_{t \geq 0}$ be Brownian motion with its natural augmented filtration. For $0 \leq u \leq t$,

$W_t = W_u + (W_t - W_u).$

The increment

$W_t - W_u$

is independent of $\mathcal{F}_u$ and satisfies

$W_t - W_u \sim N(0, t - u).$

Therefore,

$E[W_t - W_u | \mathcal{F}_u] = 0,$

and

$E[(W_t - W_u)^2 | \mathcal{F}_u] = t - u.$

Table 1: Comparison of the three recognition routes.						
Route		Applicable form	Required check	Logical conclusion	Principal advantage	Principal limitation
I.	Conditional-expectation definition	Any adapted integrable process	$E(X_t \mathcal{F}_u) = X_u, 0 \leq u \leq t$	True martingale	General; closest to the definition	Conditional expectations may be difficult
II.	Itô drift inspection	$dX_t = a_t dt + b_t dW_t$	$a_t = 0$, followed by suitable integrability	Local martingale first; true martingale after added conditions	Fast and natural for Itô processes	Zero drift alone need not imply a true martingale
III.	Square-integrable stochastic-integral theorem	$X_t = X_0 + \int_0^t H_s dW_s$	H predictable and $E \int_0^T H_s^2 ds < \infty$	Square-integrable martingale	Direct theorem with moment formulas	Applies only to stochastic-integral form
Note: Route I is the formal definition and is the most general method. Route II is a fast diagnostic for Itô processes, but “zero drift” should first be interpreted as a local-martingale result unless suitable integrability is also verified. Route III is not a new definition of martingale; it is a specialized theorem for processes of the form of a Brownian stochastic integral. It should not be applied directly to ordinary time integrals such as $\int_0^t W_s ds$.						

These two identities generate many of the direct proofs below.

For linear processes, the first conditional moment is often sufficient. For nonlinear processes, higher conditional moments appear. Squaring the Brownian increment decomposition gives

$$W_t^2 = W_u^2 + 2W_u(W_t - W_u) + (W_t - W_u)^2.$$

Taking conditional expectations yields

$$E[W_t^2 | \mathcal{F}_u] = W_u^2 + (t - u).$$

The extra term

$$t - u$$

is the predictable conditional growth of squared Brownian motion. Subtracting its accumulated form t produces the compensated process $W_t^2 - t$.

The direct route is especially valuable in validation, because it reveals the precise conditional error. It does not merely state that a drift exists; it identifies how the expected future increment depends on the current state.

2.3 Route II: The Itô drift test

Suppose X admits the decomposition

$$X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s.$$

In differential notation,

$$dX_t = a_t dt + b_t dW_t.$$

The first integral is a predictable finite-variation component. The second is a continuous local-martingale component, assuming the stochastic integral is locally well defined.

If

$$a_t \neq 0,$$

the process normally contains predictable conditional accumulation and is not a local martingale.

If

$$a_t = 0,$$

then

$$X_t = X_0 + \int_0^t b_s dW_s$$

is a local martingale under the relevant local integrability assumptions.

The word *local* cannot simply be discarded.

Definition 1: Local martingale

An adapted process X is a local martingale if there exists an increasing sequence of stopping times

$$\tau_n \uparrow \infty$$

almost surely such that every stopped process

$$X_{t \wedge \tau_n}$$

is a martingale.

Local martingales are designed to permit processes that behave like martingales before they reach increasingly remote boundaries. They are fundamental in stochastic calculus because Itô integrals are naturally constructed locally.

A true martingale requires global integrability and the unconditional validity of

$$E[X_t | \mathcal{F}_u] = X_u.$$

A local martingale need not satisfy this equality globally.

Definition 2: Strict local martingale

A strict local martingale is a local martingale that is not a true martingale.

This distinction is economically meaningful. A nonnegative local martingale is a supermartingale, so

$$E[X_t] \leq E[X_0].$$

If the inequality is strict for some horizon, the process is a strict local martingale. In asset-pricing models, a nonnegative discounted price that is only a strict local martingale may display a gap between its current value and the conditional expectation of its future value. Such models are connected to mathematical representations of asset-price bubbles, option-pricing anomalies and failures of standard valuation identities.

A convenient sufficient condition for upgrading a driftless Brownian integral to a true martingale is

$$E\left[\int_0^T b_s^2 ds\right] < \infty.$$

Under this condition,

$$\int_0^t b_s dW_s$$

is square-integrable and hence a true martingale on $[0, T]$.

Thus, the practical logic is:

zero drift $\Rightarrow$ local martingale,

while

zero drift + suitable integrability $\Rightarrow$ true martingale.

For

$$X_t = f(t, W_t),$$

Itô's formula gives

$$df(t, W_t) = \left(f_t \left| \frac{1}{2} f_{ww} \right| dt + f_w dW_t\right).$$

The local-martingale condition is therefore

$$f_t + \frac{1}{2} f_{ww} = 0.$$



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This is the backward heat equation for Brownian motion. Its satisfaction removes the Itô drift, but an integrability argument is still needed when a true-martingale result is required.

2.4 Route III: The square-integrable stochastic-integral theorem

The third route applies to the specialized representation

$$M_t = M_0 + \int_0^t H_s dW_s.$$

It does not apply merely because a process is square-integrable. It applies because the process is a Brownian stochastic integral whose integrand satisfies a particular L^2 condition.

Theorem 1: Square-integrable Brownian stochastic integrals

Let H be predictable process³ and assume that, for every finite T ,

$$E\left[\int_0^T H_s^2 ds\right] < \infty.$$

Then

$$M_t = \int_0^t H_s dW_s$$

is a square-integrable martingale. Moreover,

$$E[M_t] = 0$$

and

$$E[M_t^2] = E\left[\int_0^t H_s^2 ds\right].$$

Proof

The Itô integral is initially defined for simple predictable processes and extended through L^2 completion.

Itô isometry gives

$$E[M_t^2] = E\left[\int_0^t H_s^2 ds\right] < \infty.$$

Thus

$$M_t \in L^2,$$

which also implies

$$M_t \in L^1.$$

For $u \leq t$,

$$M_t = M_u + \int_u^t H_s dW_s.$$

The future stochastic-integral increment has conditional mean zero:

$$E\left[\int_u^t H_s dW_s \mid \mathcal{F}_u\right] = 0.$$

Therefore,

$$E[M_t \mid \mathcal{F}_u] = M_u.$$

Hence M is a square-integrable martingale.

The theorem provides more than the statement “there is no drift.” It verifies that the integral is integrable, identifies its second moment and upgrades the local-martingale structure to a true martingale.

However, the theorem must not be applied directly to

$$\int_0^t H_s ds.$$

An ordinary time integral is a finite-variation process, not a Brownian stochastic integral.

Figure 1 summarizes the three-route architecture.

For a quant, the efficient workflow is not to select one route permanently. It is to use the routes sequentially:

definition $\rightarrow$ structural decomposition $\rightarrow$ integrability verification.

3 Canonical processes and the errors they expose

Table 2 previews the processes studied in this section.

3.1 Brownian motion

Let

$$X_t = W_t.$$

Since

$$W_t \approx N(0, t),$$

we have

$$E[|W_t|] < \infty.$$

Proof by conditional expectation

For $u \leq t$,

$$W_t = W_u + (W_t - W_u).$$

Therefore,

$$\begin{aligned} E[W_t \mid \mathcal{F}_u] &= E[W_u + (W_t - W_u) \mid \mathcal{F}_u] \\ &= W_u + E[W_t - W_u \mid \mathcal{F}_u] \\ &= W_u. \end{aligned}$$

Hence, W_t is a martingale.

Proof by the drift test

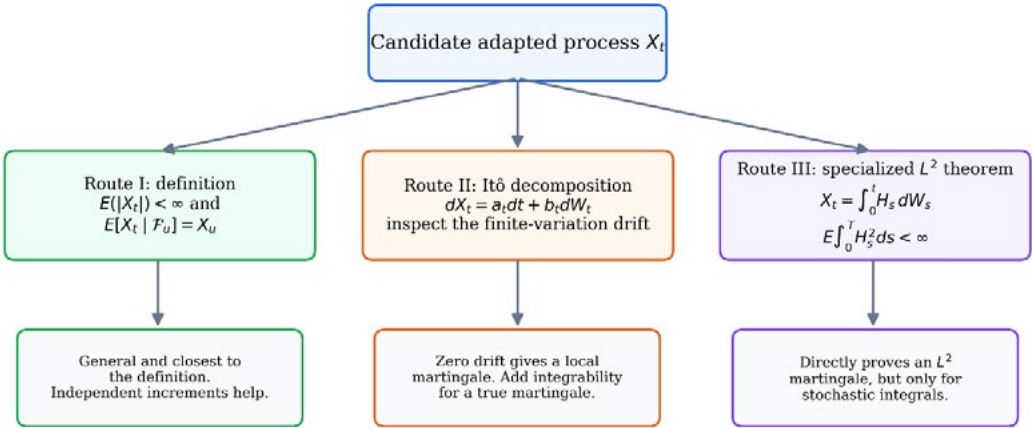
The differential is

$$dX_t = dW_t.$$

There is no dt -component. Moreover,

$$E\left[\int_0^T 1^2 ds\right] = T < \infty.$$

Figure 1: Three routes for recognizing Brownian Martingales. Route I is fully general. Route II identifies the finite-variation drift but requires care when moving from a local to a true martingale. Route III directly establishes a square-integrable martingale only for appropriately integrable Brownian stochastic integrals.



Triangulate whenever possible: related routes, different logical roles.

Table 2: Classification of canonical Brownian processes.				
Process	Classification	Conditional-expectation result	Itô or drift result	Preferred route
W_t	Martingale	$E_u[W_t] = W_u$	$dW_t = dW_t$	I or II
W_t^2	Submartingale, not martingale	$E_u[W_t^2] = W_u^2 + t - u$	$d(W_t^2) = dt + 2W_t dW_t$	I and II
$W_t^2 - t$	Martingale	$E_u[W_t^2 - t] = W_u^2 - u$	$d(W_t^2 - t) = 2W_t dW_t$	I, II, or III

Thus,

$$W_t = \int_0^t 1 dW_s$$

is a square-integrable martingale.

Practical interpretation

A realized Brownian path may exhibit prolonged increases or declines. Nothing in the martingale definition requires visual flatness or frequent mean reversion.

This matters when inspecting simulated pricing paths. A trending path is not evidence against a martingale. The relevant question is whether, conditional on present information, the expected future increment is zero.

3.2 Squared Brownian motion

Consider

$$X_t = W_t^2.$$

The unconditional expectation is

$$E[W_t^2] = t.$$

Its conditional growth is even more informative.

Proof by conditional expectation

Using

$$W_t = W_u + (W_t - W_u),$$

we obtain

$$W_t^2 = W_u^2 + 2W_u(W_t - W_u) + (W_t - W_u)^2.$$

Taking conditional expectations,

$$\begin{aligned} E[W_t^2 | \mathcal{F}_u] &= W_u^2 + 2W_u E[W_t - W_u | \mathcal{F}_u] + E[(W_t - W_u)^2 | \mathcal{F}_u] \\ &= W_u^2 + 0 + (t - u) \\ &= W_u^2 + (t - u). \end{aligned}$$

If $t > u$,

$$E[W_t^2 | \mathcal{F}_u] > W_u^2.$$

Therefore W_t^2 is not a martingale. It is a submartingale.



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Proof by the drift test

For

$$f(w) = w^2,$$

we have

$$f'(w) = 2w$$

and

$$f''(w) = 2.$$

Itô's formula gives

$$d(W_t^2) = 2W_t dW_t + \frac{1}{2}(2)dt,$$

so

$$d(W_t^2) = dt + 2W_t dW_t.$$

The positive dt -term corresponds exactly to the conditional increment $t - u$.

Practical interpretation

The example is the simplest demonstration that nonlinear transformations create drift through quadratic variation.

A quant who applies an ordinary chain rule to W_t^2 might incorrectly write

$$d(W_t^2) = 2W_t dW_t$$

and miss the term

$$dt.$$

In a pricing or risk model, the same omission can lead to an incorrect drift, an invalid hedge or a mistaken martingale claim.

3.3 Compensated squared Brownian motion

Define

$$M_t = W_t^2 - t.$$

This subtracts the predictable quadratic growth of W_t^2 .

Proof by conditional expectation

From the preceding result,

$$E[W_t^2 | \mathcal{F}_u] = W_u^2 + (t - u).$$

Thus,

$$\begin{aligned} E[M_t | \mathcal{F}_u] &= E[W_t^2 - t | \mathcal{F}_u] \\ &= W_u^2 + (t - u) - t \\ &= W_u^2 - u \\ &= M_u. \end{aligned}$$

Therefore M_t is a martingale.

Proof by the drift test

Using

$$d(W_t^2) = dt + 2W_t dW_t,$$

we obtain

$$\begin{aligned} d(W_t^2 - t) &= d(W_t^2) - dt \\ &= dt + 2W_t dW_t - dt \\ &= 2W_t dW_t. \end{aligned}$$

The drift cancels.

Proof by the square-integrable stochastic-integral theorem

Integrating the differential gives

$$W_t^2 - t = 2 \int_0^t W_s dW_s.$$

The integrand satisfies

$$\begin{aligned} E\left[\int_0^T (|2W|)^2 ds\right] &= 4 \int_0^T E[W_s^2] ds \\ &= 4 \int_0^T s ds \\ &= 2T^2 < \infty. \end{aligned}$$

Hence $W_t^2 - t$ is a square-integrable martingale.

Practical interpretation

Compensation is one of the most reusable ideas in quantitative finance. A process is decomposed into:

predictable finite variation + martingale innovation.

The same logic appears in:

- discounted asset-price dynamics;
- compensated jump processes;
- default-intensity models;
- likelihood-ratio and density processes;
- realized-variance decompositions; and
- drift adjustments under a pricing measure.

Figure 2 compares W_t , W_t^2 and $W_t^2 - t$ along the same simulated Brownian path.

4 Conclusion

This first part has separated three related but logically distinct routes for recognizing a martingale. The conditional-expectation definition is decisive and fully general. The Itô drift test is faster for continuous-time models, but zero drift initially identifies a local martingale. The square-integrable stochastic-integral theorem supplies a true-martingale conclusion only after its predictable-integrand and integrability assumptions have been verified.

The Brownian examples show how these routes reinforce one another. Brownian motion is a martingale; squared Brownian motion is a submartingale because

Figure 2. One Brownian path and two nonlinear transformations. The three curves use an identical Brownian shock history. The subtraction of t does not remove randomness; it removes the predictable accumulation in the conditional second moment.



quadratic variation creates predictable growth; and subtracting that growth produces a square-integrable martingale. For pricing and model validation, the efficient habit is therefore to specify the measure and filtration, identify the finite-variation component, and then verify the integrability required by the intended conclusion.

Part II turns to a more deceptive class of errors: stochastic integrals, ordinary time integrals and processes that share unconditional moments while having different conditional dynamics.

**This article is part of the author's three-part series on heuristic martingale material in Wilmott Magazine, together with Part II: Stochastic Integrals Are Not Time Integrals and Part III: From Exponential Martingales to Pricing Measures.*

Code Availability

A reproducible Python Jupyter notebook covering the full martingale series (Parts I–III) is available at GitHub repository: https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/Wilmott_quantitative_finance_heuristic_material_martingale. Readers can reproduce Figures 1 and 2 and the analytical classifications reported in Tables 1 and 2. No external dataset is required.

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Endnotes

1. A reproducible Python Jupyter notebook covering the full martingale series (Parts I–III) is available at GitHub repository: https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/Wilmott_quantitative_finance_heuristic_material_martingale
2. Parts II and III appear as separate articles in *Wilmott Magazine*.
3. H denotes a single predictable process defined independently of the terminal time T . The condition 'for every finite T ' only requires that the same process H be square-integrable on every finite time interval. It does not permit the integrand itself to change with T . Therefore, an expression such as $\int_0^t (t-u) dw_u$ cannot be treated directly as $\int_0^t H_u dw_u$ under this theorem, because its integrand depends explicitly on the terminal time t . Actually, $\int_0^t (t-u) dw_u$ is not a martingale, and its proof is detailed in the next article in this three-part series, *Martingales in Practice, Part II: Stochastic Integrals Are Not Time Integrals*.



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